

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO -2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior.
(L3)

Assessment Tool : Design Task

Weightage: 20%

QUESTIONS: Total Marks: 20

Question 1: Responsive Navigation Bar Design - Marks(4)

Suppose that an e-commerce website requires a responsive navigation bar that adapts across desktop, tablet, and mobile devices.

(a) Identify key components required in a navigation bar for an e-commerce website.

A navigation bar is an important part of a website because it helps users move between different pages easily.

Key Components

1. Website Logo
2. Home Link
3. Products/Categories
4. Search Bar

(b) Design a responsive navigation bar using appropriate CSS techniques.

```
<!DOCTYPE html>
<html>
<head>
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<link rel="stylesheet" href="style.css">
</head>

<body>

<nav class="navbar">

<div class="logo">
ShopEasy
</div>

<ul class="nav-links">
<li><a href="#">Home</a></li>
<li><a href="#">Products</a></li>
<li><a href="#">Categories</a></li>
<li><a href="#">Contact</a></li>
</ul>

</nav>

</body>
</html>
```

(c) Demonstrate how the navigation adapts to smaller screens (e.g., hamburger menu).

```
*{
margin:0;
padding:0;
box-sizing:border-box;
}

.navbar{
display:flex;
justify-content:space-between;
align-items:center;
background:#333;
padding:15px;
}
```

```
.logo{  
color:white;  
font-size:24px;  
font-weight:bold;  
}
```

```
.nav-links{  
display:flex;  
list-style:none;  
}
```

```
.nav-links li{  
margin-left:20px;  
}
```

```
.nav-links a{  
color:white;  
text-decoration:none;  
font-size:18px;  
}
```

```
.nav-links a:hover{  
color:yellow;  
}
```

(d) Evaluate the usability and suggest improvements for better user experience.

Evaluation

The responsive navigation bar works efficiently on desktop, tablet, and mobile devices.

Advantages

- Easy navigation.
- Better user experience.
- Saves screen space on mobile.
- Quick access to important pages.
- Clean and organized layout.

Question 2: Product Hover Effect - Marks(4)

Suppose that an e-commerce website wants to enhance user interaction through visual effects

on product items.

(a) Identify suitable CSS properties for hover effects.

Hover effects improve the appearance of product cards and make the website more interactive. When the user moves the mouse over a product, CSS hover properties change its style.

```
.card:hover{  
    background-color:#f5f5f5;  
}
```

(b) Suggest appropriate CSS animation or transition for product Interaction.

CSS transitions and animations make hover effects smooth instead of changing instantly.

```
.card{  
    transition:0.3s ease;  
}  
.card:hover{  
    transform:scale(1.05);  
}
```

(c) Design a hover effect for a product card.

```
<!DOCTYPE html>  
<html>  
<head>  
<link rel="stylesheet" href="style.css">  
</head>  
  
<body>  
  
<div class="card">  
  
  
  
<h2>Wireless Headphones</h2>  
  
<p>₹2,999</p>  
  
<button>Buy Now</button>  
  
</div>  
  
</body>  
</html>
```

(d) Evaluate performance and user experience considerations.

Evaluation

The hover effect improves the visual appeal of the e-commerce website and makes products more engaging.

Performance Considerations

1. Use lightweight CSS animations.
2. Avoid excessive shadows and complex animations.
3. Optimize product images for faster loading.
4. Use hardware-accelerated properties such as **transform** and **opacity**.
5. Test animations on different browsers and devices.

Question 3: Layout Selection (Flexbox vs Grid) - Marks(4)

Suppose that a dashboard layout needs to be designed for a web application.

(a) Identify layout requirements of a dashboard (rows, columns, alignment).

Answer:

A dashboard is a web page that displays multiple types of information, such as statistics, charts, reports, and user details, in an organized manner.

Layout Requirements

1. Header

Displays the application title, logo, notifications, and user profile.

(b) Compare Flexbox and Grid based on layout needs.

Flexbox

- Arranges items in one direction (horizontal or vertical).
- Suitable for navigation bars and small layouts.

CSS Grid

- Arranges items in both rows and columns.
- Suitable for complex layouts such as dashboards.

(c) Choose the most suitable layout technique and justify your choice.

```
<div class="dashboard">
```

```
<div class="header">Header</div>
```

```
<div class="sidebar">Sidebar</div>
```

```
<div class="content">Main Content</div>
```

```
<div class="footer">Footer</div>
```

```
</div>
```

```
<div class="dashboard">
```

```
<div class="header">Header</div>
```

```
<div class="sidebar">Sidebar</div>
```

```
<div class="content">Main Content</div>
```

```
<div class="footer">Footer</div>
```

```
</div>
```

(d) Evaluate the scalability and responsiveness of the chosen approach.

Scalability

- CSS Grid allows developers to add new dashboard widgets without changing the overall structure.
- Additional rows and columns can be added easily.
- Suitable for both small and large dashboards.

Advantages

- Adapts to desktop, tablet, and mobile devices.
- Maintains proper spacing and alignment.
- Easy to customize.
- Improves user experience.
- Reduces layout issues on different screen sizes.

Question 4: Mobile-First Blog Layout - Marks(4)

Suppose that a blog page must be designed following a mobile-first approach.

(a) Identify key sections of a blog layout.

A mobile-first blog layout is designed for mobile devices first and then enhanced for larger screens using CSS media queries.

Key Sections of a Blog Layout

1. Header
 - Displays the blog title and logo.
2. Navigation Menu
 - Contains links such as Home, About, Categories, and Contact.

(b) Design a mobile-first layout using CSS.

```
<!DOCTYPE html>
<html>
<head>
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<link rel="stylesheet" href="style.css">
</head>

<body>

<header>
<h1>My Blog</h1>
</header>

<nav>
<a href="#">Home</a>
<a href="#">About</a>
<a href="#">Articles</a>
<a href="#">Contact</a>
</nav>

<div class="container">

<main>
<h2>Latest Blog Post</h2>
```

```
<p>Welcome to my blog website.</p>
</main>
```

```
<aside>
<h3>Recent Posts</h3>
<ul>
<li>HTML Basics</li>
<li>CSS Tutorial</li>
<li>JavaScript Guide</li>
</ul>
</aside>
```

```
</div>
```

```
<footer>
Copyright © 2026
</footer>
```

```
</body>
</html>
```

(c) Demonstrate how the layout scales for larger screens using media queries.

```
@media screen and (min-width:768px){
```

```
  nav{
    flex-direction:row;
    justify-content:center;
  }
```

```
  .container{
    flex-direction:row;
    gap:20px;
  }
```

```
  main{
    flex:3;
  }
```

```
  aside{
    flex:1;
  }
```

```
}
```


(d) Evaluate accessibility and readability of the design.

Accessibility

1. Use semantic HTML elements (**header**, **nav**, **main**, **aside**, **footer**).
2. Ensure sufficient color contrast between text and background.
3. Use readable font sizes (at least 16px).
4. Add **alt** text for images.
5. Make links and buttons keyboard accessible.
6. Provide enough spacing between elements for touch devices.

Question 5: JavaScript Event Handling for Dynamic Menu - Marks(4)

Suppose that a website includes a dynamic menu that responds to user interactions.

(a) Identify appropriate JavaScript events for menu interaction.

JavaScript events are actions performed by the user or browser that trigger specific functions. For a dynamic menu, the following events are commonly used.

Important JavaScript Events

1. **click**
 - Triggered when the user clicks a menu or button.
 - Used to open or close the navigation menu.

(b) Design event handling logic for menu toggle functionality.

Logic

1. Create a menu button (**☰**).
2. Initially hide the navigation menu on mobile devices.
3. When the user clicks the menu button, display the menu.
4. If the menu is already open, hide it again.

```
<button id="menuBtn">☰ Menu</button>
```

```
<ul id="menu">
  <li><a href="#">Home</a></li>
  <li><a href="#">Products</a></li>
  <li><a href="#">About</a></li>
  <li><a href="#">Contact</a></li>
```


(c) Implement interaction for user actions (click/hover).

```
const menuBtn = document.getElementById("menuBtn");  
const menu = document.getElementById("menu");
```

```
menuBtn.addEventListener("click", function(){
```

```
    if(menu.style.display === "block"){  
        menu.style.display = "none";  
    }  
    else{  
        menu.style.display = "block";  
    }  
}
```

```
});
```

(d) Evaluate performance and suggest improvements.

Performance Evaluation

The dynamic menu provides a smooth and responsive navigation experience. It improves usability by allowing users to access menu items quickly without reloading the page.

Advantages

- Faster navigation.
- Interactive user interface.
- Easy to implement and maintain.
- Reduces page clutter on mobile devices.
- Works well on desktops, tablets, and smartphones.

Code of Conduct:

I Ch. Bhargav Ram (192425334) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: